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MILITARY SPECIFICATION

PROJECTILE, TRAINING, 8 INCH: M423, SHIPPING ASSEMBLY

This specification is mandatory for use by the activities of the Army Munitions Command.

1. SCOPE

1.1 Scope. This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Training Projectile 8 Inch: M423 Shipping Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-C-50565(MU) - Container, Shipping and Storage Atomic Projectile: M500.

STANDARDS

MILITARY

AN6227-67 - Packing Preformed.
BFBX1-141196, 443769 - Pin, Dowel.
BXSX2-585892 - Pin, Spring.

FSC 1110

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MS-9048-034 - Pin, Spring.
MS-35672-20 - Pin, Grooved.

DRAWINGS

ORDNANCE CORPS

8787101 - Plug, Nose Atomic Projectile.
8787102 - Ogive Atomic Projectile.
8787113 - Nut, Plain, Round.
8787145 - Spring, Helical Compression.
8787146 - Cup, Spring.
8787152 - Blank, Obturating Band.
8787176 - Pin, Roll.
8787179 - Base Jacket, Ring Assembly, Training.
8787181 - Base, Training.
8787184 - Base, Training, Atomic Projectile.
8787185 - Nut Base Plug Training.
8787186 - Sleeve, Base Plug, Training.
8787188 - Body Section, Training, Atomic Projectile Rear.
8787189 - Training Body Nut Assembly.
8787190 - Body Sleeve Assembly, Training.
8787194 - Body Section, Training, Atomic Projectile Front.
8787197 - Gun Tube, Atomic Weapon, Training.
8787198 - Tube, Body, Training.
8787199 - Sleeve, Tube, Training.
8787239 - Stop, Nuclear Target.
8787240 - Bushing, Machine, Thread.
8787868 - Projectile, Training, 8 Inch, M423 Shipping Assembly.
8794333 - Spacer, Sleeve.
8795632 - Dashpot, Dummy, Training.
8795634 - Dashpot, Atomic Weapon, Dummy.
8795639 - Gasket.
8795678 - Spacer.
8795689 - Container, Metal, Ammo, for Proj 8 Inch, M423 Shipping Assembly, Pkg.

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8804300 - Pin, Shoulder, Headless
8804302 - Screw, Machine
8805718 - Packing, Preformed
8808270 - Cylinder, Training
8808568 - Sleeve
8808569 - Ring
8830814 - Base Plug and Tube, Assembly, Training
8837260 - Pin, Dowel
9209432 - Stop, Ring
9209782 - Grommet Assembly
9221216 - Dashpot Assembly, Atomic Weapon, Modified, Training
ET-8787868 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Shipping Assembly shall comply with all requirements specified on Drawing (dwg) 8787868, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Paint adhesion (see dwg 8787868). Paint adhesion, of the shipping assembly shall comply with Paint Adhesion, section 3, TT-C-490 when tested as specified in 4.4.1

3.4 Serialization. The contractor shall assign serial numbers as required by drawings.

3.4.1 Unclassified items. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.4.2 Classified items. Production quantities shall not be revealed. Random serial numbers shall be selected by the contractor in a manner which has been approved by the procuring agency. These numbers shall not be duplicated.

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

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3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilty for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product Acceptance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed, as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Ring (see dwg 8808569) covering a detail of dwg 8787190.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Inside diameter, max	Gage
	102. Length, min	Gage
	103. Material (see 3.1)	Visual
	104. Objective evidence (see 6.2 for protective finish in accordance with dwg 8808569)	Visual
Minor:		
	201. Surface finish	Visual
	202. Serialization (see 3.4)	Visual
	203. Stamp	Visual
	204. Workmanship (see 3.5)	Visual/Manual

4.3.2.1.2 Sleeve (see dwg 8808568) covering a detail of dwg 8787190.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Large outside diameter	Gage
	102. Width of wrench slot	Gage
	103. Length from shoulder to right face	Gage

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Major (cont):
104. Small inside diameter.....Gage
105. Small outside diameter, min.....Gage
106. Distance across wrench slot flats.....Gage
107. Total length.....Gage
108. Material (see 3.1).....Visual
109. Objective evidence (see 6.2) for protective finish in accordance with dwg 8808568.....Visual

Minor:
201. Surface finish.....Visual
202. Serialization (see 3.4).....Visual
203. Stamping.....Visual
204. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.3 Body sleeve assembly training (see dwg 8787190 covering a detail of dwg 8787188.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of inside thread, max.	Gage
	102. Minor diameter of inside thread, max.	Gage
	103. Pitch diameter of outside thread	Gage
	104. Major diameter of outside thread	Gage
	105. Diameter of outside thread clearance	Gage
	106. Width of outside thread clearance, max.	Gage
	107. Diameter of dowel pin holes before assembly, max.	Gage
	108. Steel exposed surfaces covered with lubricant	Visual
	109. Aluminum exposed surfaces coated	Visual
	110. Objective evidence (see 6.2) for pin, dowel in accordance with dwg 8837260	Visual
Minor:		
	201. Stamping	Visual
	202. Serialization (see 3.4)	Visual
	203. Workmanship (see 3.5)	Visual/Manual

4.3.2.1.4 Spacer (see dwg 8795678 and 8795679 (see dwg 8787678)covering a detail of dwg 9209782.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Inside diameter.....Gage
102. Material (see 3.1).....Visual

Minor:
201. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.5 Grommet assembly (see dwg 9209782 covering a detail of dwg 8787188).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Spacers tight in body	Manual
102.	Material (see 3.1)	Visual
103.	Staples flush with inside contour if staples are used	Manual
104.	Nylon core present	Visual
Minor:	None defined.	

4.3.2.1.6 Training body nut assembly (see dwg 8787189) covering a detail of dwg 8787188.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Outside diameter	Gage
102.	Diameter of left chamfer	Gage
103.	Diameter of thread clearance	Gage
104.	Diameter adjacent to thread clearance	Gage
105.	Pitch diameter inside thread	Gage
106.	Minor diameter inside thread	Gage
107.	Distance from left face to thread clearance	Gage
108.	Distance from left face to right side of thread clearance	Gage
109.	Distance to left datum diameter	Gage
110.	Distance to right datum diameter	Gage
111.	Diameter of outside thread clearance	Gage
112.	Right inside diameter	Gage
113.	Pitch diameter of outside thread	Gage

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Major (cont):

114.	Major diameter of outside thread	Gage
115.	Width of outside thread clearance, max.	Gage
116.	Distance from right face to shoulder	Gage
117.	Total length	Gage
118.	Contour of rotating band	Visual
119.	Steel surfaces coated	Visual
120.	Material (see 3.1)	Visual

Minor:

201.	Surface finish	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.7 Stop, nuclear target (see dwg 8787239) covering a detail of dwg 8787188.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

Major:

101.	Distance from left face to right shoulder	Gage
102.	Distance from left face to left shoulder	Gage
103.	Smallest diameter	Gage
104.	Middle diameter	Gage
105.	Location of smallest diameter	Gage
106.	Radius and chamfer present	Visual
107.	Largest diameter	Gage
108.	Width of slot	Gage
109.	Depth of slot	Gage
110.	Total length	Gage
111.	Material (see 3.1)	Visual

Minor:

201.	Surface finish	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.8 Bushing, machine thread (see dwg 8787240) covering a detail of dwg 8787188.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Inside diameter.....Gage
102. Pitch diameter of thread, min.....Gage
103. Major diameter of thread, min.....Gage
104. Depth of slot.....Gage
105. Total length.....Gage
106. Width of slot.....Gage
107. Material (see 3.1).....Visual
108. Objective evidence (see 6.2) for protective finish in accordance with dwg 8787240.....Visual

Minor:
201. Surface finish.....Visual
202. Stamping.....Visual
203. Serialization (see 3.4).....Visual

4.3.2.1.9 Stop, ring (see dwg 9209432) covering a detail of dwg 8787188.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Inside diameter, max.	Gage
	102. Material (see 3.1)	Visual
	103. Part coated with corrosion preventative	Visual
Minor:		
	201. Workmanship (see 3.5)	Visual/Manual

4.3.2.1.10 Body section, training, atomic projectile rear (see dwg 8787188) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of 4 dash pot holes	Gage
	102. Location of 4 nuclear target stop holes	Gage
	103. Stop welded	Visual
	104. Threads coated before assembly	Visual
	105. Diameter of small hole in dash pot hole	Gage
	106. Diameter of nuclear stop hole	Gage

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Major (cont):

107.	Depth of threaded cavity nuclear stop hole	Gage
108.	Length of thread nuclear stop hole, min	Gage
109.	Pitch diameter of thread nuclear stop hole, max	Gage
110.	Minor diameter of thread nuclear stop hole, max	Gage
111.	Large diameter dash pot hole outside	Gage
112.	Small diameter dash pot hole	Gage
113.	Large diameter dash pot hole inside	Gage
114.	Depth of large dash pot hole outside	Gage
115.	Distance from outside curvature to left side of small diameter of dash pot hole	Gage
116.	Chamfer present on inside of small diameter of dash pot hole	Visual
117.	Objective evidence (see 6.2) for pin in accordance with BFBX1-141196	Visual
118.	Pin loose after assembly	Manual
119.	No break through of pin hole	Visual
120.	Location of pin	Gage
121.	Location of stop	Gage
122.	Location of nuclear stops	Gage
123.	Total length, max	Gage
124.	Pin flush	Visual/Manual
125.	Bushing staked 2 places nuclear stop	Visual
126.	Material (see 3.1)	Visual
127.	Nuclear stops properly assembled	Visual

Minor:

201.	Painting no bare spot	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.11 Cylinder, Training (see dwg 8808270) covering a detail of dwg 8787194.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Depth of large inside diameter	Gage
102.	Large inside diameter	Gage
103.	Small inside diameter	Gage

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Major (cont):

104.	Angle of chamfer between large and small inside diameters	Gage
105.	Length of small left outside diameter	Gage
106.	Depth of cavity	Gage
107.	Pitch diameter of large threaded holes	Gage
108.	Minor diameter of large threaded holes	Gage
109.	Location of thread in large threaded holes	Gage
110.	Length of thread of large threaded holes	Gage
111.	Diameter of large threaded holes	Gage
112.	Location of large threaded holes	Gage
113.	Depth of chamfer on large threaded holes, min	Gage
114.	Diameter of small inside cavity	Gage
115.	Depth of small inside cavity	Gage
116.	Location of small threaded holes	Gage
117.	Length of right outside shoulder	Gage
118.	Total length	Gage
119.	Outside diameter	Gage
120.	Material (see 3.1)	Visual
121.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8808270	Visual

Minor:

201.	Stamping	Visual
202.	Serialization (see 3.4)	Visual
203.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.12 Body section, training, atomic projectile front (see dwg 8787194) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Pitch diameter of left thread	Gage
102.	Minor diameter of left thread	Gage
103.	Length of engagement left thread, min	Gage
104.	Left outside diameter	Gage
105.	Depth of thread clearance left thread	Gage
106.	Location of pins	Gage
107.	Width of wrench slot	Gage
108.	Location of wrench slot	Gage
109.	Pitch diameter of right thread	Gage

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Major (cont):

110.	Minor diameter of right thread	Gage
111.	Length of engagement right thread	Gage
112.	Taper present right end	Visual
113.	Right outside diameter	Gage
114.	Depth of right thread clearance	Gage
115.	Depth of small cavity	Gage
116.	Total length	Gage
117.	Distance across wrench flats	Gage
118.	Pin holes not visible inside cylinder	Visual
119.	Objective evidence (see 6.2) for dowell pin in accordance with BFBX1-141196 or BFBX1-443769 as applicable	Visual

Minor:

201.	Stamping	Visual
202.	Serialization (see 3.4)	Visual
203.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.13 Dash pot atomic weapon, dummy (see dwg 8795632) covering a detail of dwg 8795634.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

Major:

101.	Diameter adjacent to thread	Gage
102.	Distance from bottom to bottom of notch, max	Gage
103.	Pitch diameter of thread	Gage
104.	Major diameter of thread	Gage
105.	Location of notch	Gage
106.	Location of circular groove	Visual
107.	Location of small hole	Gage
108.	Diameter of notched projection	Gage
109.	Outside diameter	Gage
110.	Distance from left curvature to right curvature	Gage
111.	Distance from left end of flange to right curvature	Gage
112.	Thickness of flange	Gage
113.	Depth of thread, min	Gage
114.	Configuration in right curvature	Visual
115.	Material (see 3.1)	Visual
116.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8795632	Visual

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Minor:
201. Surface finish.....Visual
202. Stamping.....Visual
203. Serialization (see 3.4).....Visual
204. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.14 Dashpot, Atomic Weapon, Dummy (see dwg 8795634) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Height of pin	Gage
102.	Pin tight	Manual
103.	Objective evidence (see 6.2) for pin in accordance with BXSX2-585892	Visual
104.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8795634	Visual
Minor:		
201.	Marking or stamping	Visual
202.	Serialization (see 3.4)	Visual
203.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.15 Dashpot assembly, atomic weapon, modified, training (see dwg 9221216-1) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of pin hole before assembly of pin	Gage
102.	Pin tight	Manual
103.	Height of pin	Gage
104.	Pin guide secured with adhesive	Visual
105.	Disc bonded to screw with adhesive	Visual
106.	Retaining screw and filling plug secured to reservoir with surface primer and sealing compound	Visual
107.	Materials (see 3.1)	Visual
108.	Objective evidence (see 6.2) for pin, spring in accordance with MS 9048-034	Visual

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Minor:
201. Marking.....Visual
202. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.16 Blank, obturating band (see dwg 8787152) covering a detail of dwg 8787179.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Material (see 3.1)	Visual
102.	Brinell hardness	Gage
Minor:	None defined.	

4.3.2.1.17 Base, training (see dwg 8787181) covering a detail of dwg 8787179.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Outside diameter of lugs	Gage
102.	Width of 4 lugs	Gage
103.	Contour of 4 lugs	Gage
104.	Radius of inside of lugs	Gage
105.	Location of 2 small holes	Gage
106.	Depth of 2 small holes, min	Gage
107.	Diameter of 2 small holes	Gage
108.	Feather edges removed as noted	Visual/Manual
109.	Location of lugs	Gage
110.	Width of band groove	Gage
111.	Length to band groove	Gage
112.	Large outside diameter	Gage
113.	Small inside diameter	Gage
114.	Chamfer present	Visual
115.	Left side of formed washer groove to right face	Gage
116.	Total length	Gage
117.	Width of formed washer groove	Gage
118.	Contour of formed washer groove	Visual
119.	Material (see 3.1)	Visual
120.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787181	Visual

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Minor:

201.	Stamping	Visual
202.	Serialization (see 3.4)	Visual
203.	Surface finish	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.18 Base jacket, ring assembly, training (see dwg 8787179) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Inside diameter of formed rubber	Gage
102.	Angle of band outside	Gage
103.	Outside diameter of band	Gage
104.	Material (see 3.1)	Gage
Minor:		
201.	Surface finish	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.19 Tube body, training (see dwg 8787198) covering a detail of dwg 8787197.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Feather edge of thread removed as noted	Visual
102.	Pitch diameter of left inside thread	Gage
103.	Minor diameter of left inside thread	Gage
104.	Length of outside thread	Gage
105.	Length of outside thread clearance	Gage
106.	Diameter of outside thread clearance	Gage
107.	Pitch diameter of outside thread	Gage
108.	Major diameter of outside thread	Gage
109.	Distance from left face to left inside shoulder	Gage
110.	Thread clearance of left inside thread, max	Gage
111.	Distance from left face to middle inside thread	Gage
112.	Distance from left face to smallest inside diameter	Gage
113.	Distance from right face to smallest inside diameter	Gage

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Major (cont):

114.	Width of middle inside thread clearance, max	Gage
115.	Distance from smallest inside diameter to largest inside diameter	Gage
116.	Distance from right face to largest inside diameter	Gage
117.	Total length of right inside thread including clearances	Gage
118.	Length of right inside thread clearance, max	Gage
119.	Length of thread clearance left of right thread, max	Gage
120.	Outside diameter	Gage
121.	Diameter of right inside thread clearances	Gage
122.	Diameter left of right inside thread, max	Gage
123.	Largest inside diameter	Gage
124.	Diameter left of largest inside diameter, max	Gage
125.	Smallest inside diameter	Gage
126.	Diameter of middle inside thread clearance	Gage
127.	Diameter left of middle inside thread	Gage
128.	Diameter of left inside thread clearance	Gage
129.	Pitch diameter of right inside thread	Gage
130.	Minor diameter of right inside thread	Gage
131.	Pitch diameter of middle inside thread	Gage
132.	Minor diameter of middle inside thread	Gage
133.	Total length	Gage
134.	Material (see 3.1)	Visual
135.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787198	Visual

Minor:

201.	Surface finish	Visual
202.	Serialization (see 3.4)	Visual
203.	Stamping	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.20 Sleeve, tube, training (see dwg 8787199) covering a detail of dwg 9797197.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Diameter of flanges, min.....Gage
102. Inside diameter.....Gage
103. Total length.....Gage
104. Material (see 3.1).....Visual

Minor:
201. Surface finish.....Visual
202. Stamping.....Visual
203. Serialization (see 3.4).....Visual
204. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.21 Gun tube, atomic weapon, training (see dwg 8787197) covering a detail of dwg 8830184.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) for pin, dowel in accordance with BFBX1-141191..Visual
102. Pins tight.....Manual
103. Gap between tube body and tube sleeve, max.....Gage
104. No break through inside of sleeve.....Visual
105. Damaged protective finish coated with laquer.....Visual

Minor:
201. Stamping.....Visual
202. Serialization (see 3.4).....Visual
203. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.22 Nut base plug training (see dwg 8787185) covering a detail of dwg 8787184.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Location of side holes.....Gage
102. Small inside diameter.....Gage
103. Diameter of left thread clearance,max..Gage
104. Chamfer present.....Visual
105. Width of thread clearances, max.....Gage
106. Length of small inside diameter.....Gage
107. Total thickness.....Gage

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Major (cont):

108.	Pitch diameter of thread, max	Gage
109.	Minor diameter of thread, max	Gage
110.	Diameter of right thread clearance, max	Gage
111.	Outside diameter	Gage
112.	Diameter of side holes	Gage
113.	Depth of side holes, min	Gage
114.	Material (see 3.1)	Visual
115.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787185	Visual

Minor:

201.	Surface finish	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.23 Sleeve base plug, training (see dwg 8787186) covering a detail of dwg 8787184.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Location of 8 blind holes	Gage
102.	Diameter of 8 blind holes	Gage
103.	Depth of 8 blind holes, min	Gage
104.	Pitch diameter of left inside thread	Gage
105.	Minor diameter of left inside thread	Gage
106.	Feather edge removed as noted	Visual
107.	Large diameter of chamfer	Gage
108.	Diameter of left inside thread clearance	Gage
109.	Total depth of left inside thread cavity	Gage
110.	Degree of chamfer	Gage
111.	Width of left thread clearance on right inside thread, max	Gage
112.	Total length	Gage
113.	Length of right outside thread	Gage
114.	Depth of slots	Gage
115.	Depth of right thread clearance on right inside thread	Gage
116.	Pitch diameter of right outside thread	Gage
117.	Major diameter of right outside thread	Gage
118.	Feather edge of thread removed as noted	Visual
119.	Diameter of inside right thread clearances	Gage

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Major (cont):

120.	Pitch diameter of right inside thread	Gage
121.	Minor diameter of right inside thread	Gage
122.	Diameter of right outside thread clearance	Gage
123.	Width of right outside thread clearance	Gage
124.	Distance from flange to right face	Gage
125.	Total length	Gage
126.	Outside diameter	Gage
127.	Total distance across formed openings	Gage
128.	Diameter of 4 holes in right face	Gage
129.	Depth of 4 holes in right face	Gage
130.	Smallest inside diameter	Gage
131.	Width of 4 formed openings	Gage
132.	Location of 4 holes in right face	Gage
133.	Contour of formed openings	Gage
134.	Material (see 3.1)	Visual
135.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787186	Visual

Minor:

201.	Surface finish	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.24 Base, training, atomic projectile (see dwg 8787184) covering a detail of dwg 8830184.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Objective evidence (see 6.2) for pin, roll in accordance with dwg 8787176	Visual
102.	Outside diameter	Gage
103.	Diameter of pin holes	Gage
104.	Distance from bottom of pin hole to right face	Gage
105.	Total length, max	Gage
106.	Cement applied to thread before assembly	Visual
107.	Material (see 3.1)	Visual
Minor:		
201.	Stamping and marking	Visual
202.	Serialization (see 3.4)	Visual
203.	Workmanship (see 3.5)	Visual/Manual

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4.3.2.1.25 Base plug and tube assembly, training (see dwg 8830814 covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Torque on gun tube	Gage
102.	Diameter of pin holes	Gage
103.	Objective evidence (see 6.2) for pin, grooved in accordance with MS35672-20	Visual
104.	Threads coated	Visual
105.	Material (see 3.1)	Visual
Minor:		
201.	Stamping	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.26 Spacer, sleeve (see dwg 8794333) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length	*Measure
102.	Outside diameter	*Measure
103.	Inside diameter	*Measure
104.	Material (see 3.1)	Visual
105.	Objective evidence for surface finish in accordance with dwg 8794333	Visual

*Measure with suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.27 Nut, plain, round (see dwg 8787113) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length	Gage
102.	Depth of slots	Gage

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Major (cont):
- 103. Pitch diameter of thread.....Gage
- 104. Minor diameter of thread.....Gage
- 105. Width of slots.....Gage
- 106. Material (see 3.1).....Visual
- 107. Objective evidence (see 6.2) for protective finish in accordance with dwg 8787113.....Visual

Minor:
- 201. Surface finish.....Visual
- 202. Stamping.....Visual
- 203. Serialization (see 3.4).....Visual
- 204. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.28 Ogive, atomic projectile (see dwg 8787102) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Inside diameter	Gage
	102. Diameter of thread clearance	Gage
	103. Distance across wrench flats	Gage
	104. Pitch diameter of left thread	Gage
	105. Major diameter of left thread	Gage
	106. Width of wrench slots	Gage
	107. Concentricity of right end	Gage
	108. Pitch diameter of right thread	Gage
	109. Minor diameter of right thread	Gage
	110. Distance to right outside datum diameter	Gage
	111. Total length	Gage
	112. Distance to left outside datum diameter	Gage
	113. Distance to left shoulder	Gage
	114. Width of thread clearance, max	Gage
	115. Material (see 3.1)	Visual
	116. Objective evidence (see 6.2) for protective finish in accordance with dwg 8787102	Visual
Minor:		
	201. Surface finish	Visual
	202. Stamping	Visual
	203. Serialization (see 3.4)	Visual
	204. Workmanship (see 3.5)	Visual/Manual

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4.3.2.1.29 Plug, nose atomic projectile (see dwg 8787101) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length of left thread clearance	Gage
102.	Width of right thread clearance, max	Gage
103.	Length of thread, min	Gage
104.	Pitch diameter of thread	Gage
105.	Minor diameter of thread	Gage
106.	Distance to datum diameter	Gage
107.	Width of wrench slot	Gage
108.	Length from shoulder to right face	Gage
109.	Distance across wrench slot flats	Gage
110.	Material (see 3.1)	Visual
111.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787101	Visual
Minor:		
201.	Surface finish	Visual
202.	Stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.30 Gasket (see dwg 8795639) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Inside diameter	Gage
102.	Outside diameter	Gage
103.	Material (see 3.1)	Visual
104.	No tears, splits, cracks, or holes	Visual
Minor:	None defined.	

4.3.2.1.31 Screw, machine (see dwg 8804302) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101.	Width of slot	Gage
102.	Diameter of head	Gage
103.	Depth of slot	Gage
104.	Pitch diameter of thread	Gage
105.	Major diameter of thread	Gage
106.	Length of thread, min	Gage
107.	Thickness of head	Gage
108.	Length	Gage
109.	Material (see 3.1)	Visual
110.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8804302	Visual

Minor:

201.	Surface finish	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.32 Pin, shoulder, headless (see dwg 8804300) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

Major:

101.	Width of slot	Gage
102.	Pitch diameter of inside thread	Gage
103.	Minor diameter of inside thread	Gage
104.	Depth of slot	Gage
105.	Depth of inside thread, min	Gage
106.	Depth of threaded hole, max	Gage
107.	Pitch diameter of outside thread	Gage
108.	Major diameter of outside thread	Gage
109.	Outside diameter	Gage
110.	Concentricity of outside diameter	Gage
111.	Length of outside diameter	Gage
112.	Total length	Gage
113.	Material (see 3.1)	Visual
114.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8804300	Visual

Minor:

201.	Surface finish	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.33 Spring, helical compression (see dwg 8787145) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Inside diameter, min.....Gage
102. Outside diameter, max.....Gage
103. Material (see 3.1).....Visual
104. Solid height, max.....Gage
105. Right hand wind.....Visual
106. Objective evidence (see 6.2) for protective finish and removing embrittlement in accordance with dwg 8787145....Visual
107. Load at 1.90 inch, 40 lb, max.....Gage
108. Load at assembled height 22 lb, min....Gage
109. Gap at both ends.....Gage
110. Free height, max.....Gage
111. Square and ground at both ends.....Visual

Minor:
201. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.34 Cup, spring (see dwg 8787146) covering a detail of dwg 8787868.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Inside diameter.....Gage
102. Thickness of cylinder.....Gage
103. Thickness of bottom.....Gage
104. Total length.....Gage
105. Material (see 3.1).....Visual
106. Objective evidence (see 6.2) for protective finish in accordance with dwg 8787146.....Visual

Minor:
201. Surface finish.....Visual
202. Workmanship (see 3.5).....Visual/Manual

4.3.2.1.35 Projectile, training, 8 Inch, M423 Shipping assembly (see dwg 8787868) before painting.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Performed packing coated before assembly.....Visual
102. Objective evidence (see 6.2) for packing "O" ring, hydraulic in accordance with AN 6227-67.....Visual

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Major (cont):
103. Objective evidence (see 6.2) for packing performed in accordance with dwg 8805718.....Visual
104. No part missing or improperly assembled.....Visual
105. Outside diameter after painting.....Gage
106. Gap between body section, training, read body section, training, front.....Gage
107. Objective evidence (see 6.2) for protective finish in accordance with dwg 878768.....Visual
108. Material (see 3.1).....Visual

Minor: None defined.

4.3.2.1.36 Projectile, training, 8 Inch, M423 Shipping assembly (see dwg 8787868).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Paint present no bare spot.....	Visual
Minor:		
	201. Stamping or marking.....	Visual
	202. Serialization (see 3.4).....	Visual
	203. Workmanship (see 3.5).....	Visual/Manual

4.3.2.1.37 Container, metal, ammo, for proj 8 Inch, M423 shipping assy, pkg (see dwg 8795689).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence (see 6.2) for container in accordance with MIL-C-50565(MU).....	Visual
	102. No part missing or improperly assembled.....	Visual/Manual
	103. Torque on securing screws, min.....	Gage
	104. Seals properly applied.....	Visual
	105. Marking.....	Visual
	106. Projectile tight in container.....	Manual
Minor:		
	201. Workmanship (see 3.5).....	Visual/Manual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified.

Test	Test Classification
a. Paint adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Paint adhesion (see 3.3). Each shipping assembly shall be tested in accordance with 4.4.1. If any item fails to comply with 3.3, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8787868. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Paint adhesion (see 3.3). Paint adhesion shall be tested in accordance with Paint Adhesion, Section 4, TT-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

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5.1.1 Level A. The projectile shall be packaged in accordance with dwg 8795689 using the specified moisture barrier.

5.1.2 Level C. The projectile shall be packaged in accordance with dwg 8795689 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The projectile shall be packed in accordance with dwg 8795689.

5.3 Marking. Marking shall be as specified in accordance with dwg 8795689.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Identification of changes. Sinces the changes in this specification revision were so numerous, the asterisk was not used to identify the changes.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1110-A025

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OMB Approval
No. 22-R255

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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